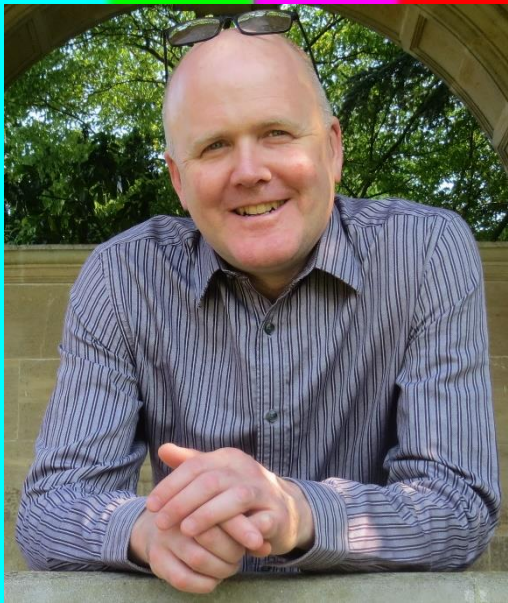
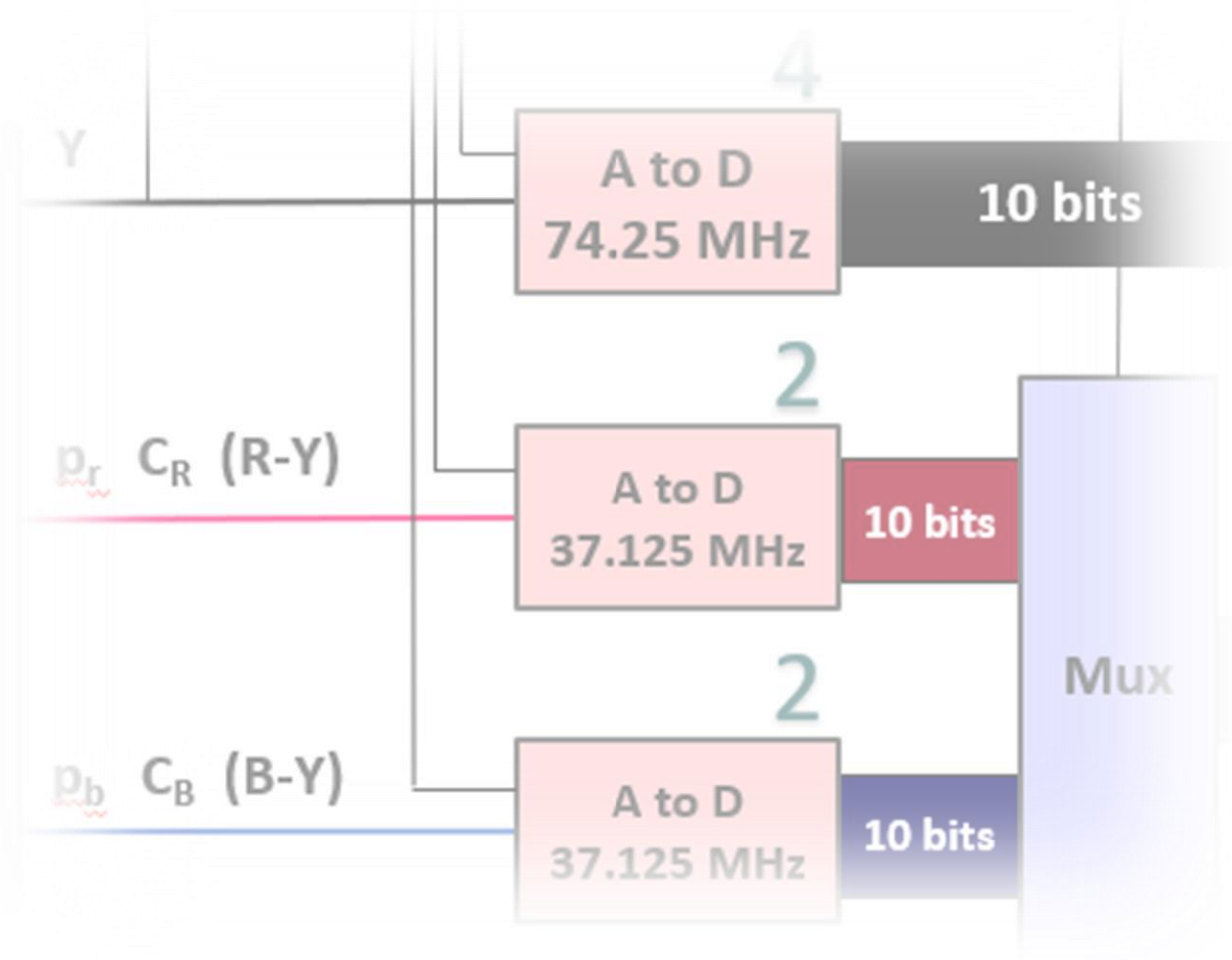


SDI Overview



Mike Dhonau



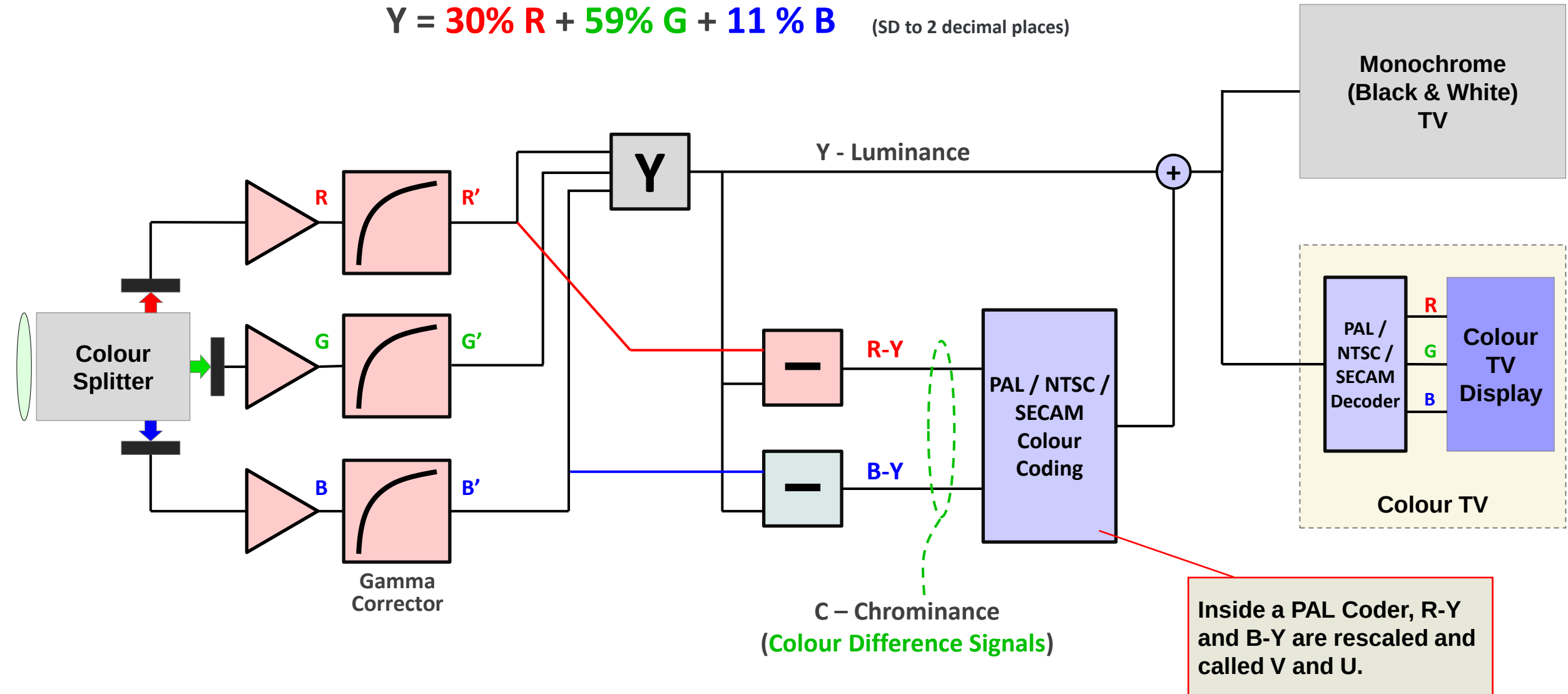
Lecture Outline

- What is SDI?
- Basic block diagram of an SDI encoder for SD and HD SDI.
 - sampling, quantising, timing reference signals, ancillary data, serialising the data, wire encoding.
- Wire Encoding
- Levels
- Timing Reference Signals
- Eye display
- 422, 411, 444, 420 Sampling
- Ancillary Data
- Standards Overview
- Cables and Connectors

- Basics of how a TV camera works.
- Pixels, lines, scanning, frames & fields, progressive and interlace.
- Line and frame rates for SD and HD.
- Bandwidth of analogue video and how it relates to resolution.
- Analogue video signals – voltages, timings, sync pulses – line and field/frame
- Gamma of displays and gamma correction (but we can do without this).
- Basics of Colour TV – How we see colour, colour mixing, RGB, luma and chroma, Y Pr Pb.

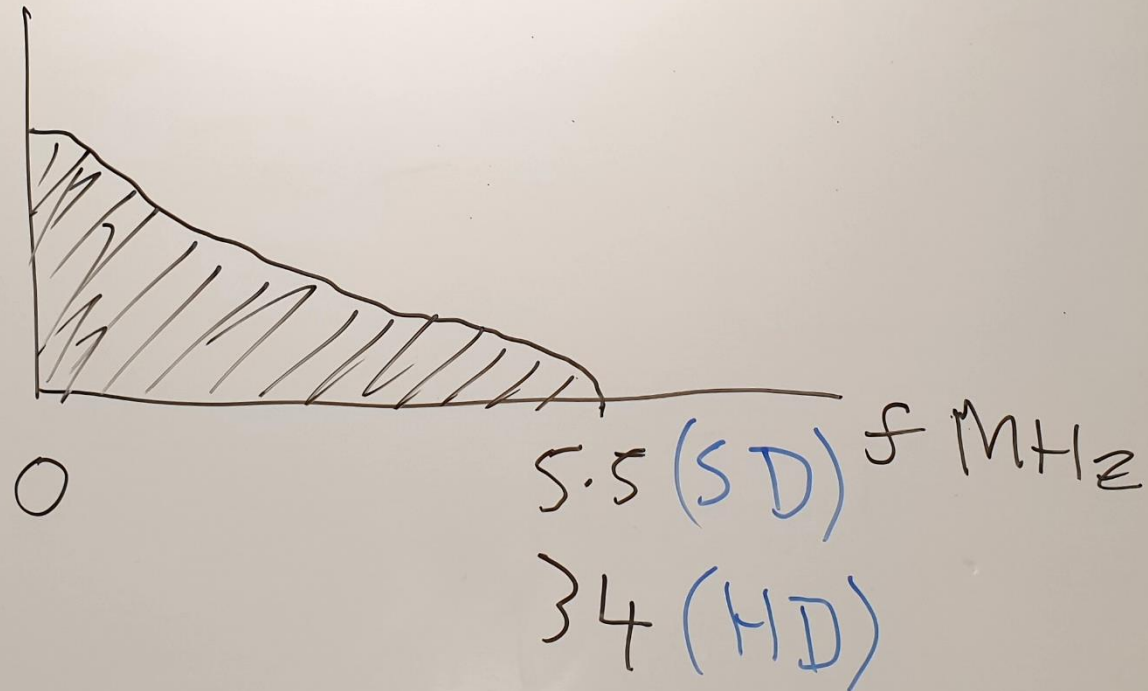
$$Y = 21\% R + 72\% G + 7\% B \quad (\text{for HD video, to 2 decimal places})$$

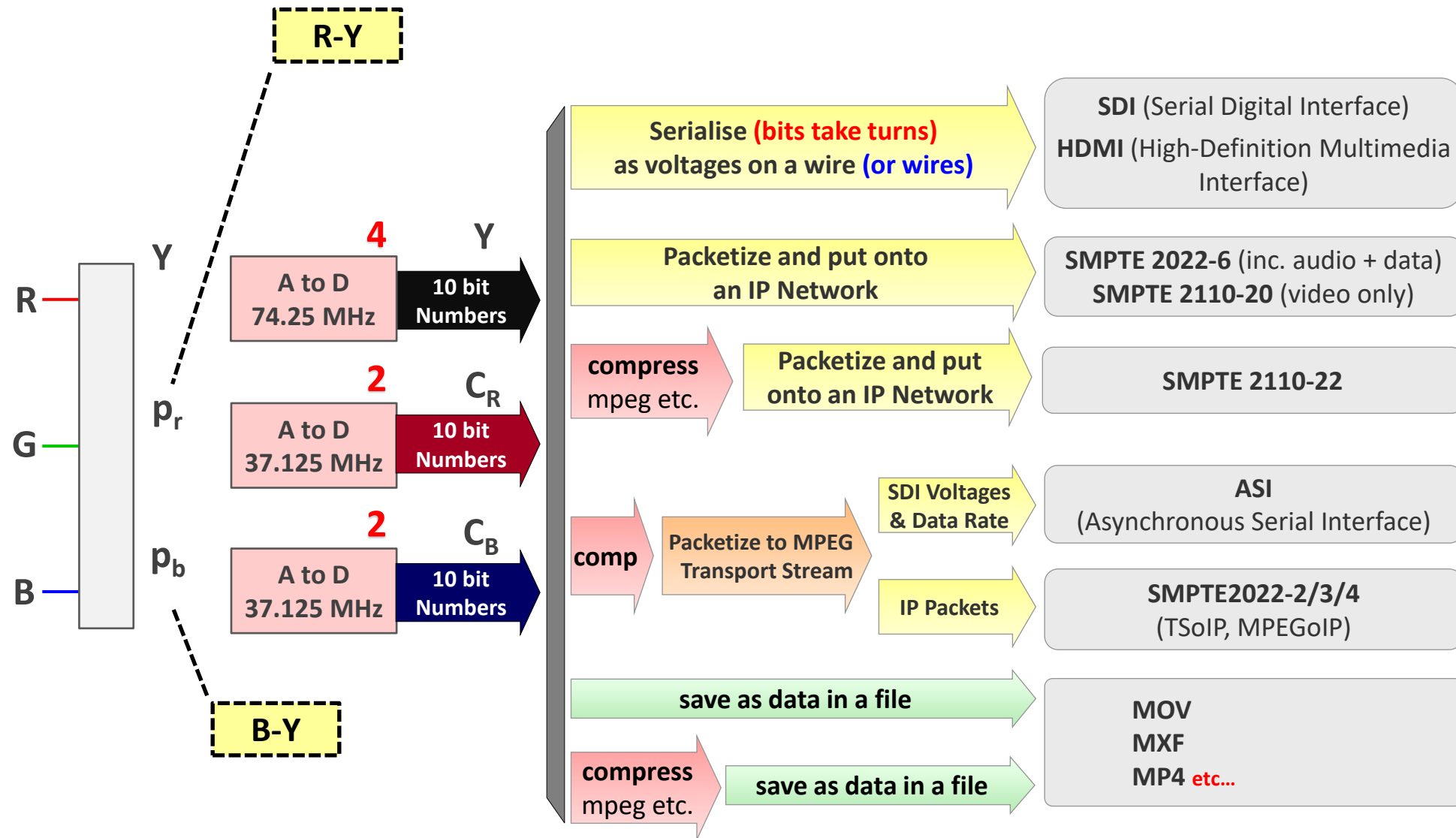
$$Y = 30\% R + 59\% G + 11\% B \quad (\text{SD to 2 decimal places})$$



Analogue Video

Bandwidth





- Small network switches support up to 144 x 100G ports.
 - Approx **4600 x 4600** for **3G**
 - Approx **1152 x 1152** for **UHD**
- Active (AOC) Direct-attach for short in-room connections.
- Multimode (OM4) for <400m.
- Single mode (OS1/OS2) for longer runs.



Single Mode Connections



AOC Using
SFP28
Connector &
Multimode
Fibre

- IP fibre from the CCU.
- IP fibre to the Multiviewers.



SDI Wiring



Camera CCUs

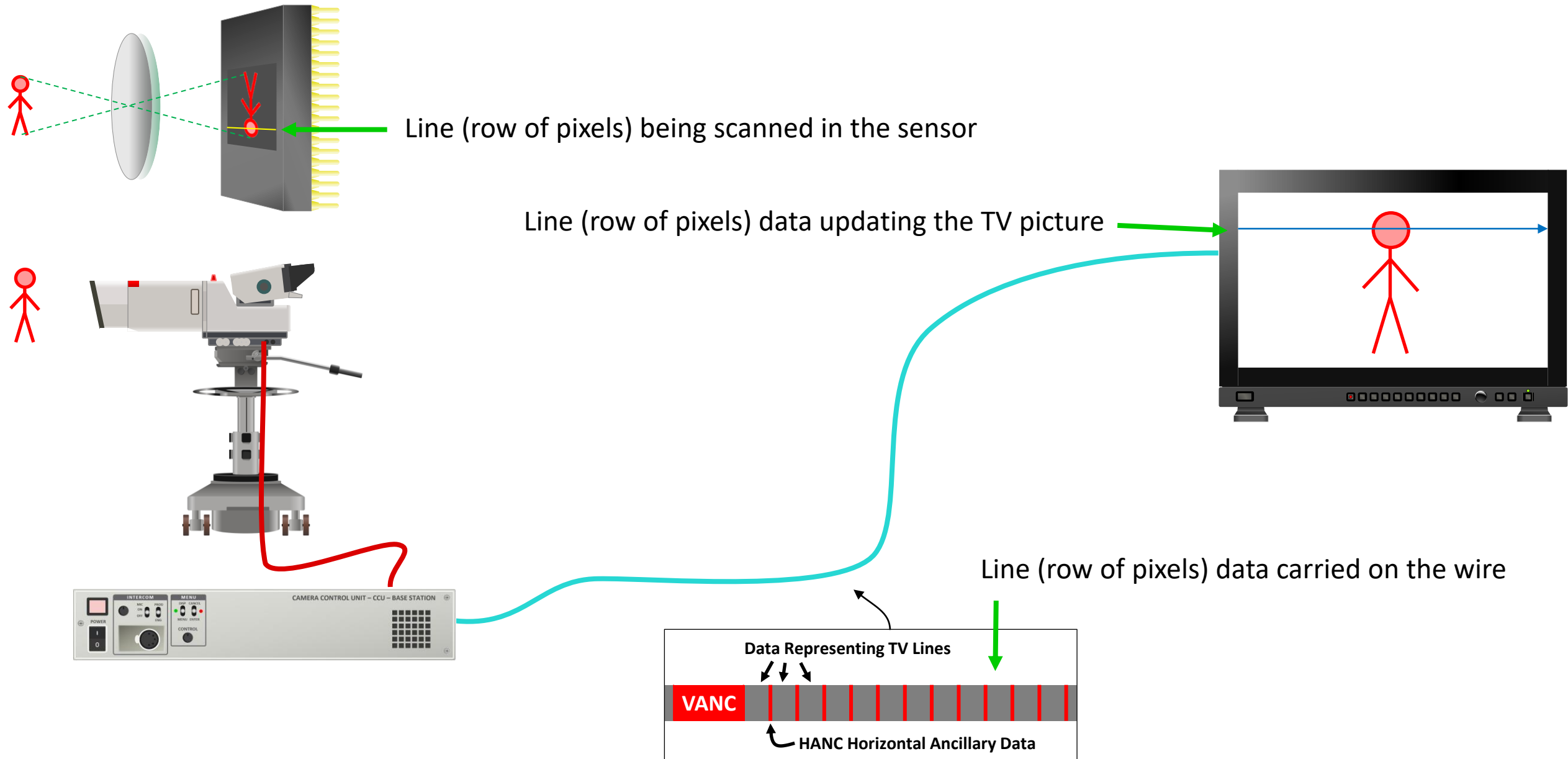


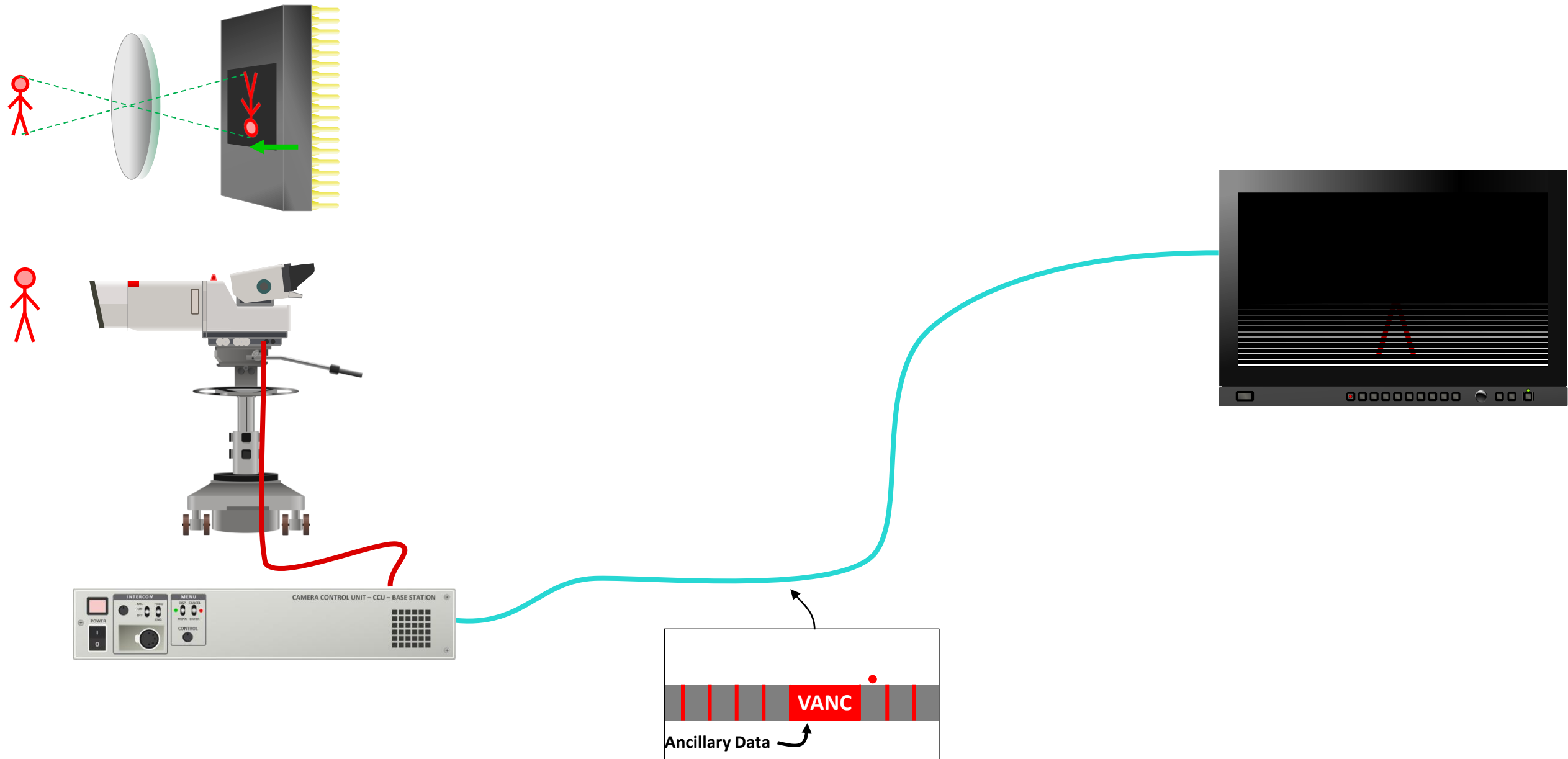
Multiviewers



Core Switches

- **LESS CABLES COMPARED TO SDI.**





SD – 270 Mbits / sec

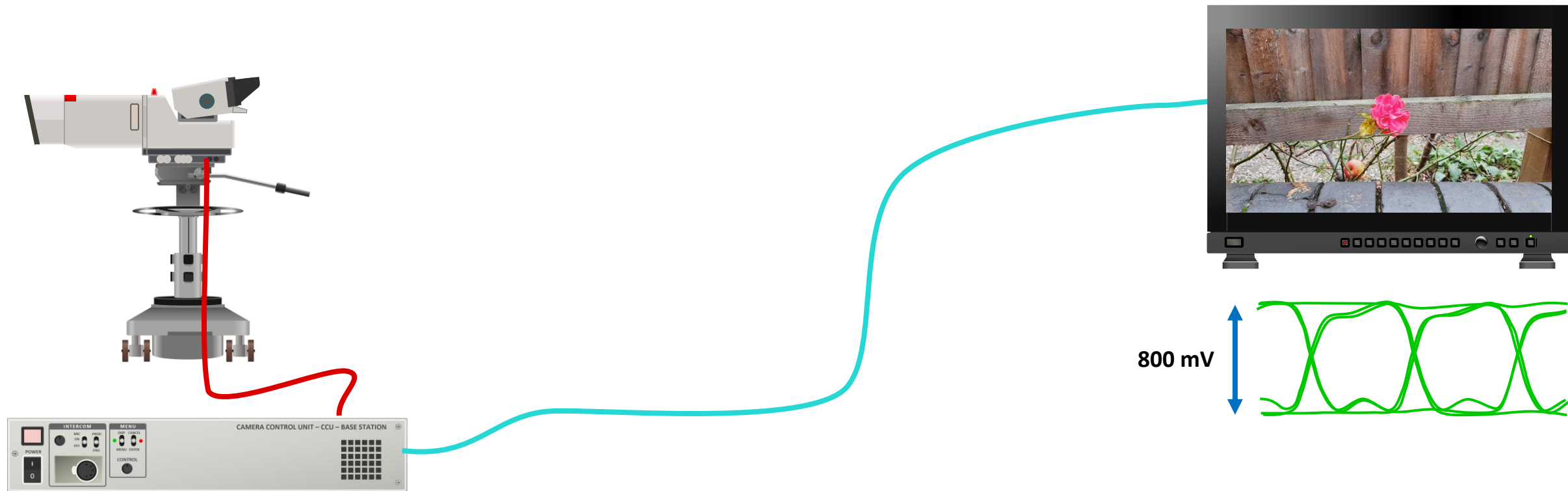
- SMPTE ST 259M
- ITU-R BT.601
- ITU-R BT.656

HD – 1.485 Gbits / sec

- SMPTE ST 292M
- ITU-R BT.709
- ITU-R BT.1120

UHD – 12 Gbits / sec

- SMPTE ST 2082
- ITU-R BT.2020
- ITU-R ITU-R BT.2077



SDI – Serial Digital Interface

SD – 270 Mbits / sec

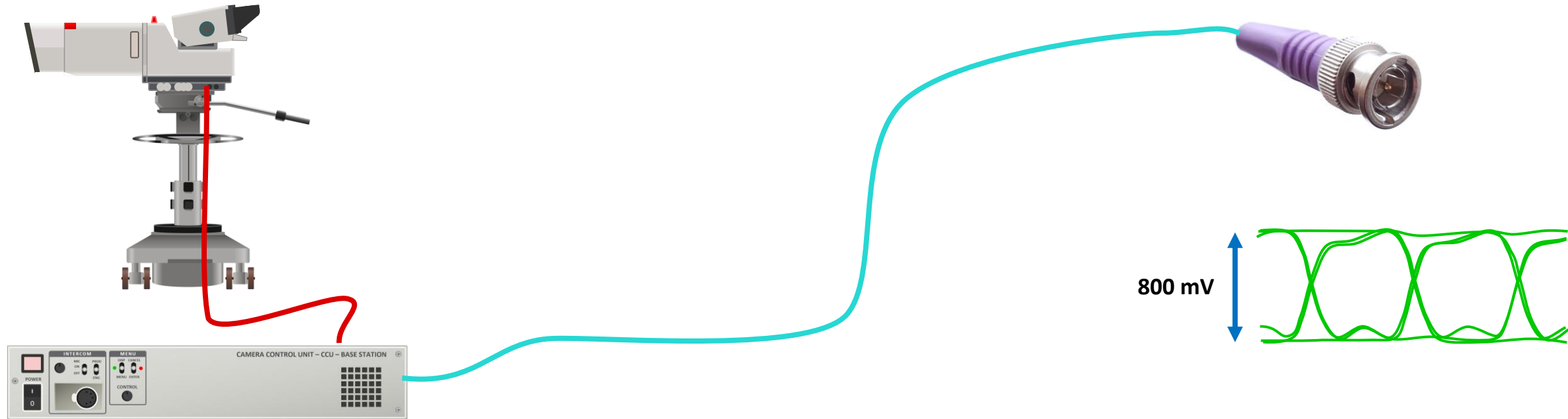
- SMPTE ST 259M
- ITU-R BT.601
- ITU-R BT.656

HD – 1.485 Gbits / sec

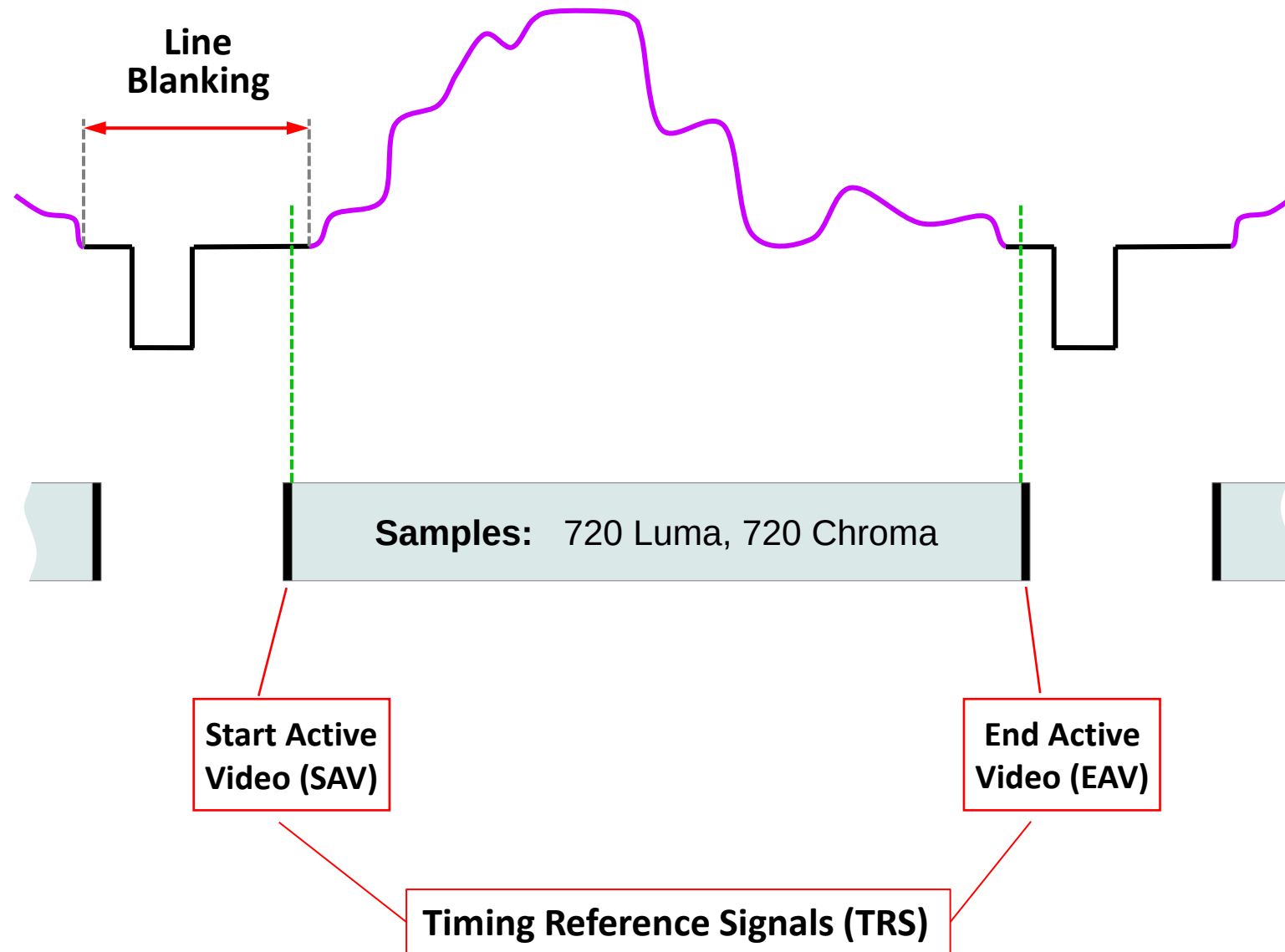
- SMPTE ST 292M
- ITU-R BT.709
- ITU-R BT.1120

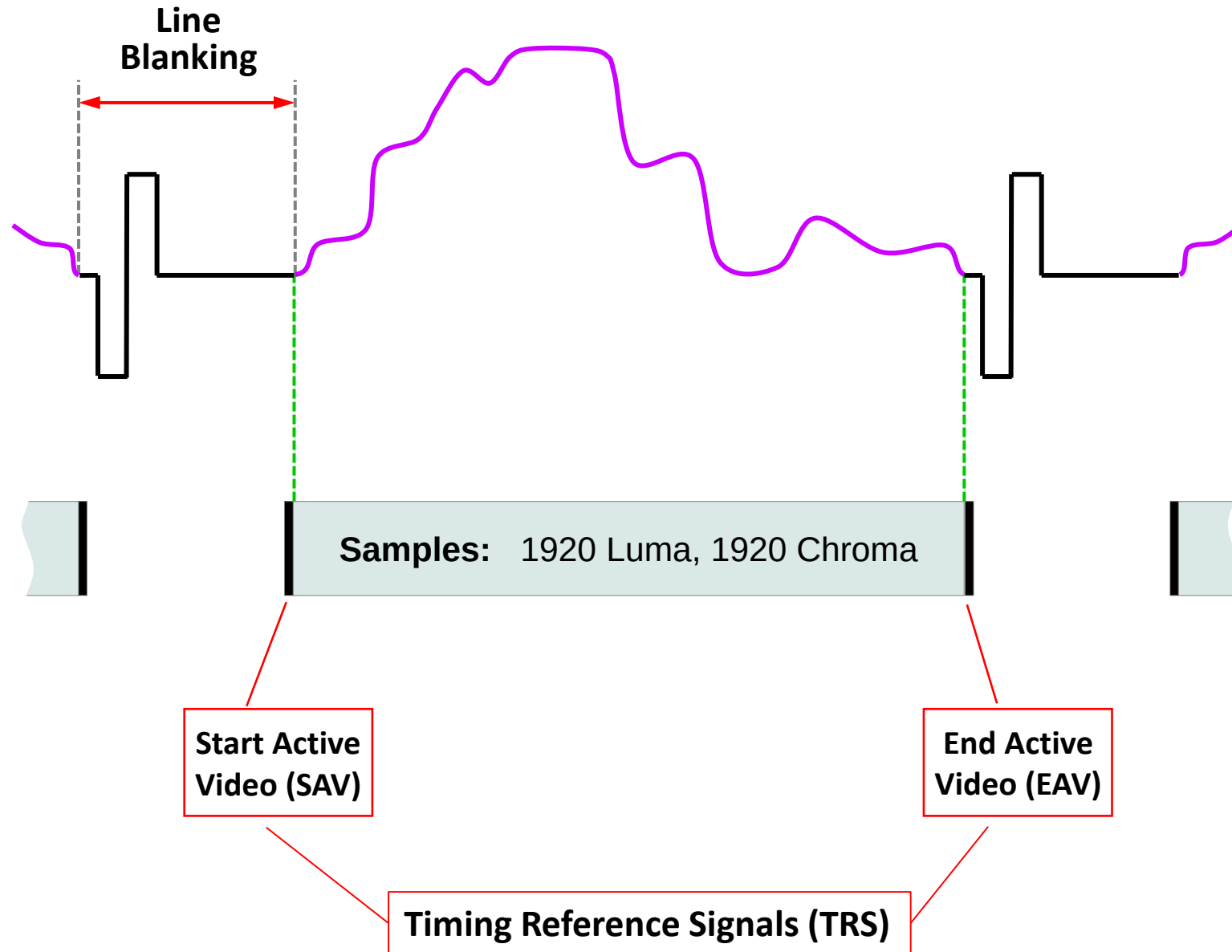
UHD – 12 Gbits / sec

- SMPTE ST 2082
- ITU-R BT.2020
- ITU-R ITU-R BT.2077



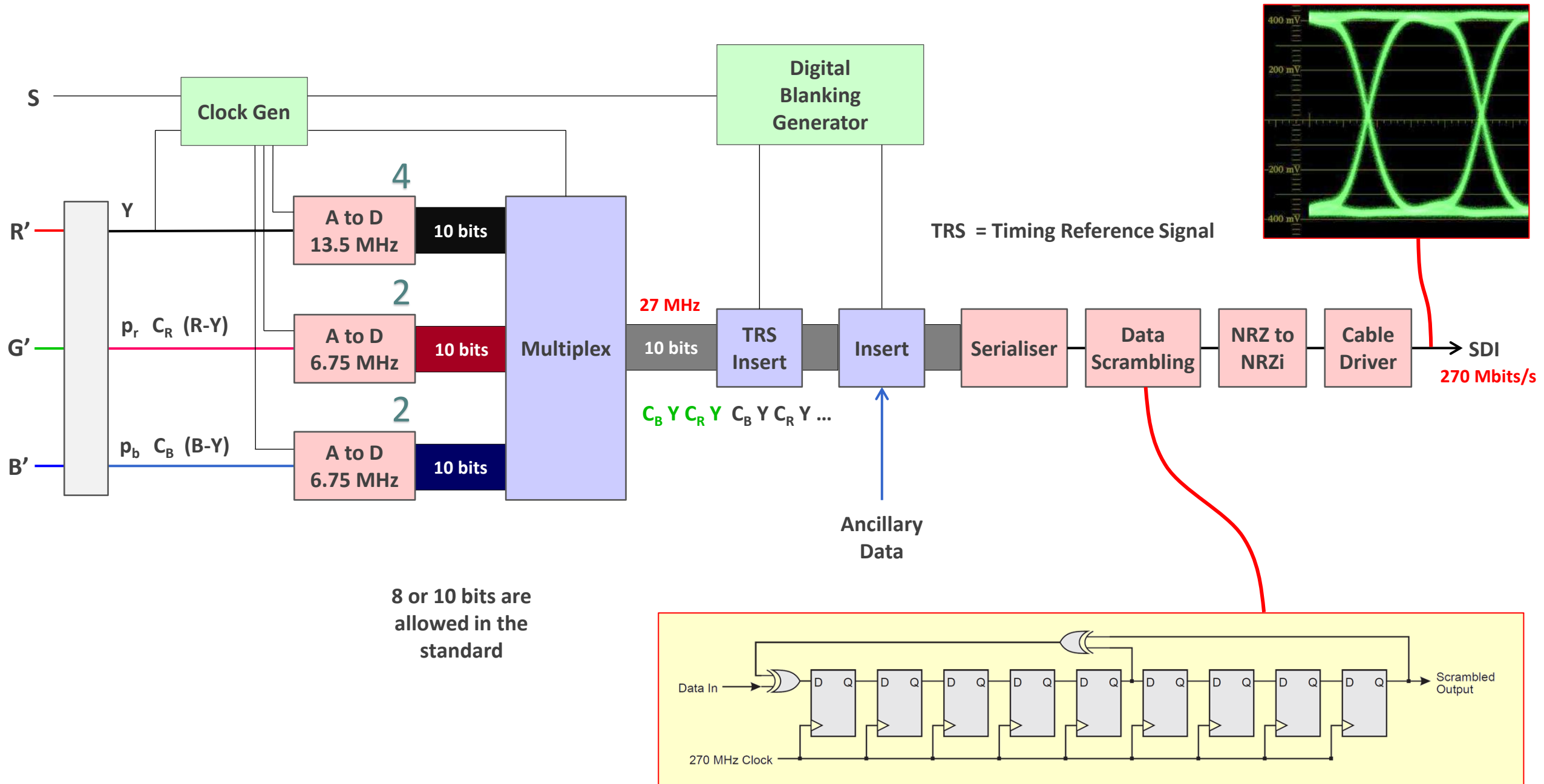
SDI – Serial Digital Interface

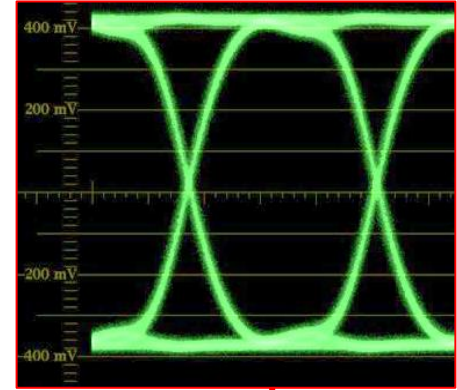




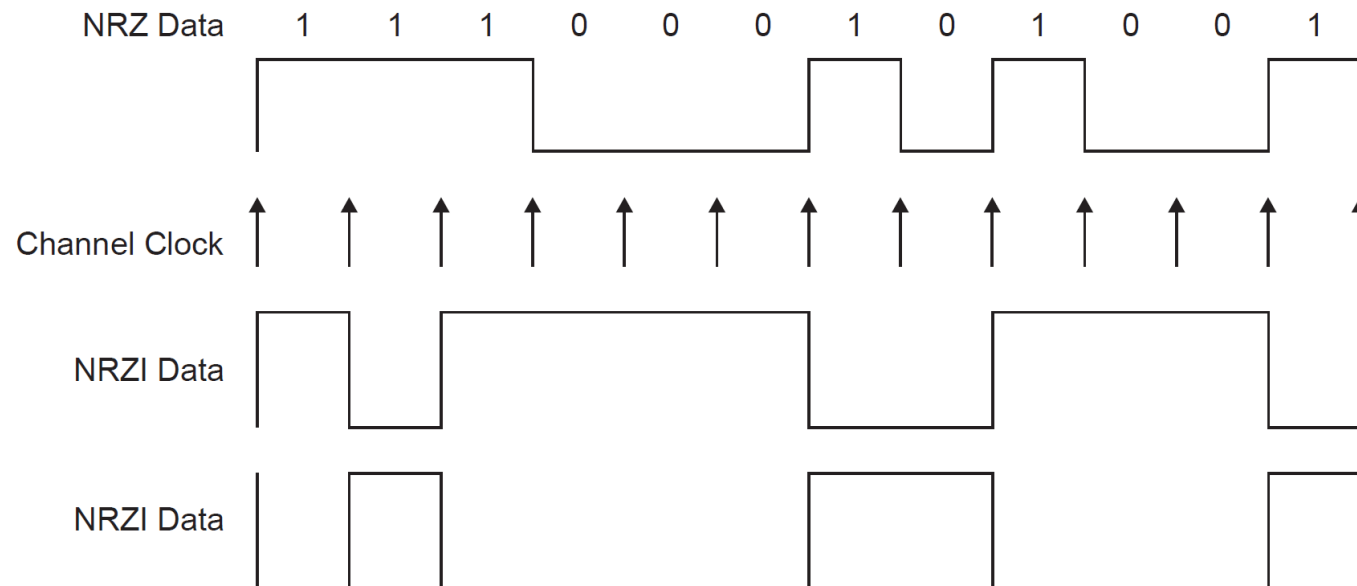
Serial Digital Interface - SD

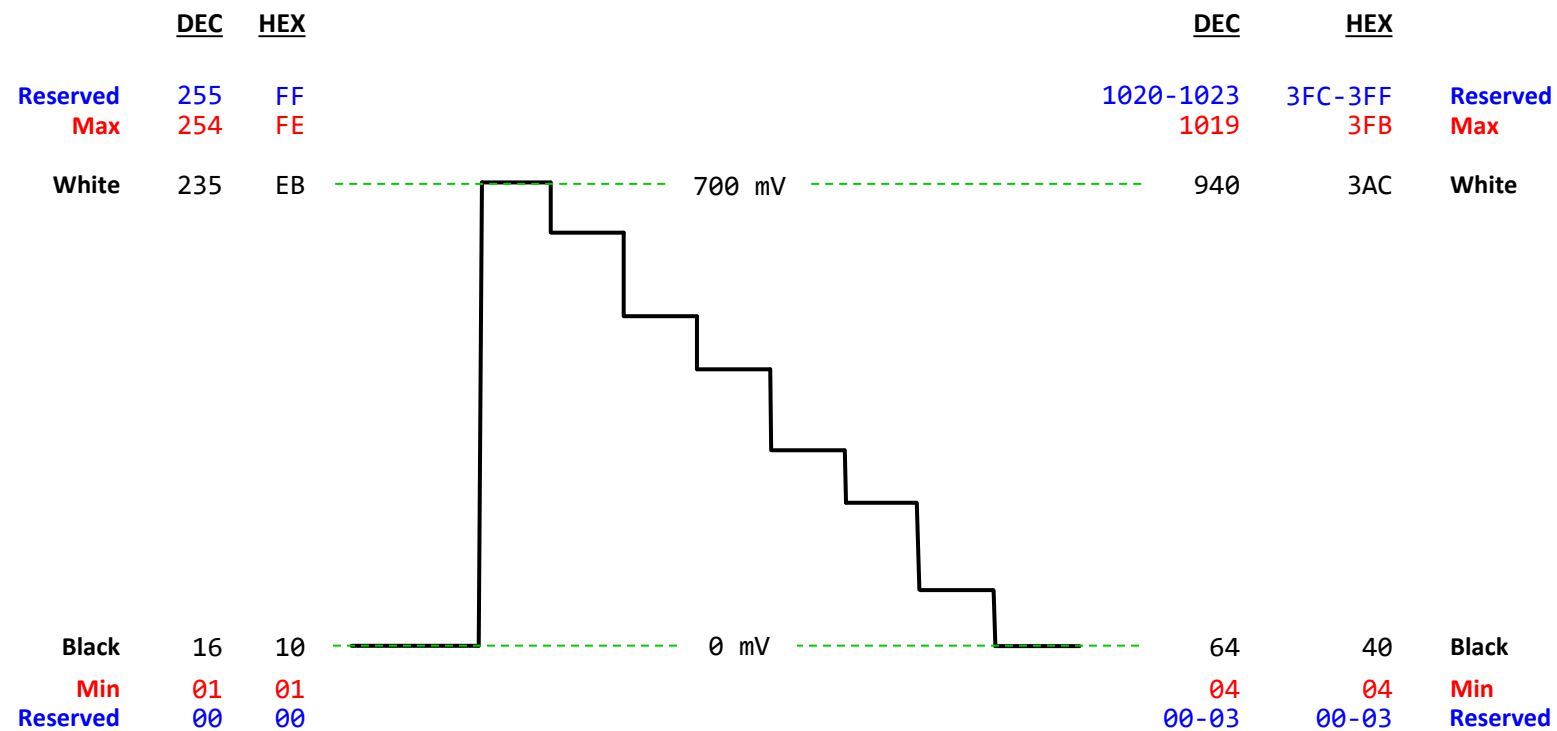
14

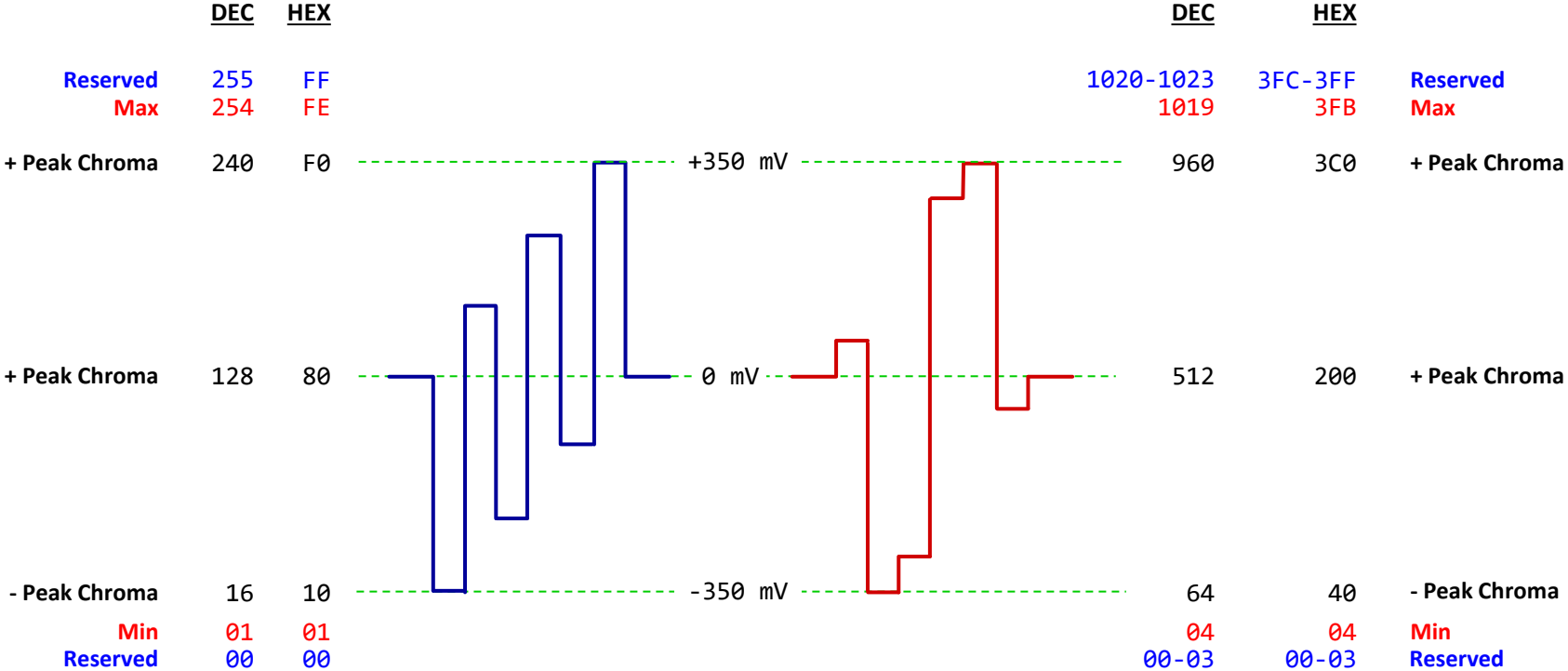




- SDI uses NRZi encoding.
 - Data = 1 change at clock edge
 - Data = 0 don't change at clock edge
- Long run of 0s means no change – not good – too much low frequency energy.
- So multiply it by self clocking pseudo-random sequence

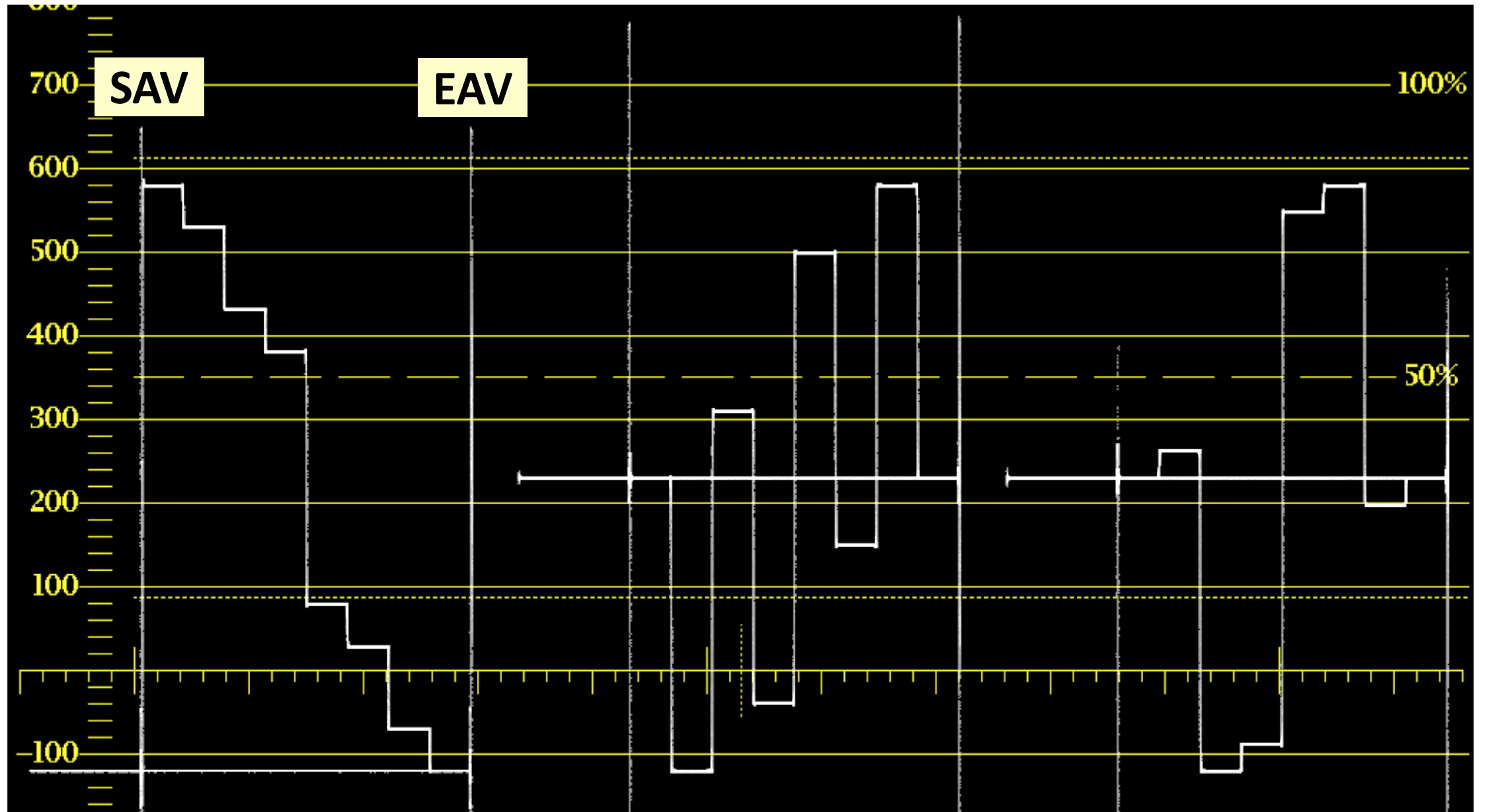


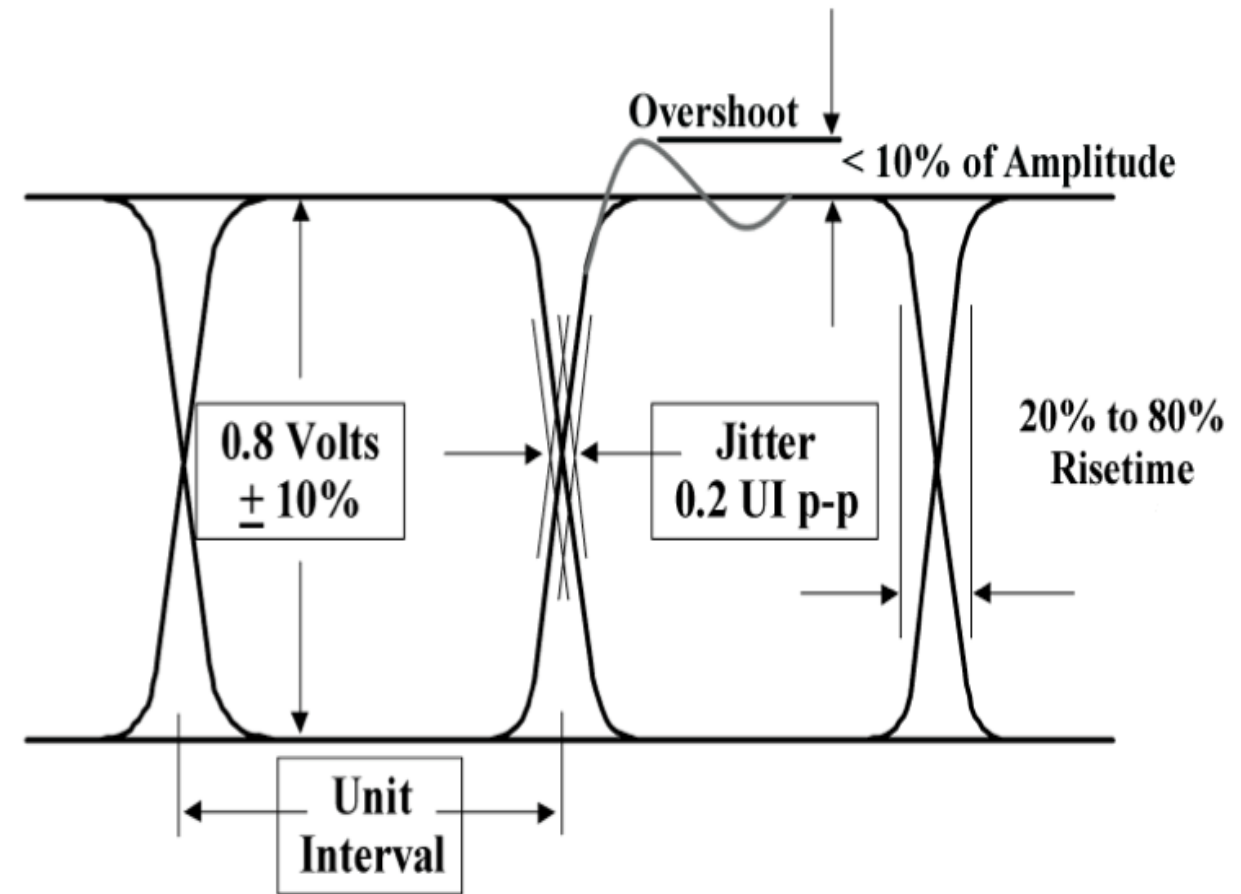




- The Timing Reference Signal (TRS) provides synchronising information.
- It does a **similar job** as an **analogue sync pulse**.
- The TRS consist of **four** 10-bit words:
 - The **first 3 words** are always the same— hex 3FF (dec 1024), 0, 0
 - These levels do not occur during active video – so the receiver can recognise the TRS.
 - The **4th word** (**XYZ word**) consists of 3 flag bits, along with an error correcting code.
- There is a TRS at both ends of the active line.
 - **SAV** – Start Active Video
 - **EAV** – End Active Video

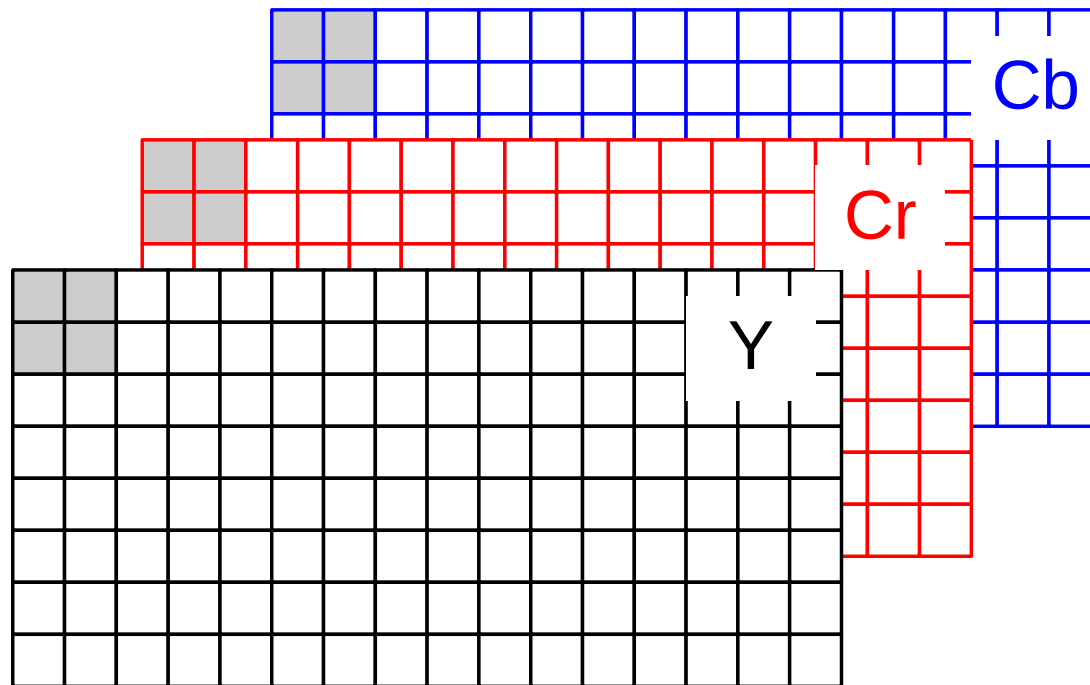
- The flags bits in the 4th word are known as **H**, **F**, and **V**.
- The TRS at the **start** of horizontal blanking (and therefore at the end of the active line) has the **H bit set to 1**.
 - This is **End of Active Video**, or **EAV**.
- The TRS appearing immediately before the start of the active video has the **H bit set to 0**;
 - This is the **Start of Active Video** or **SAV**.
- The **V bit** is used to indicate the start of the vertical blanking region.
 - An **EAV** packet with **V bit set to 1** indicates the **following line** is part of the **vertical interval**.
 - An **EAV** packet with **V bit set to 0** indicates the **following line** is part of the **active picture**.
- The **F bit** is used in interlaced and segmented-frame formats.
 - **F bit set to 0** means **field 1** or **segment 1**.
 - **F bit set to 1** means **field 2** or **segment 2**.
 - In **progressive** scan formats, the **F bit** is always set to **0**.



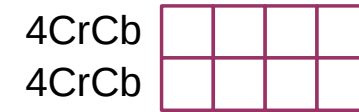
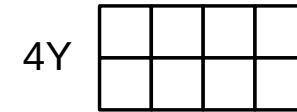


4:4:4 Video Sampling

4:4:4

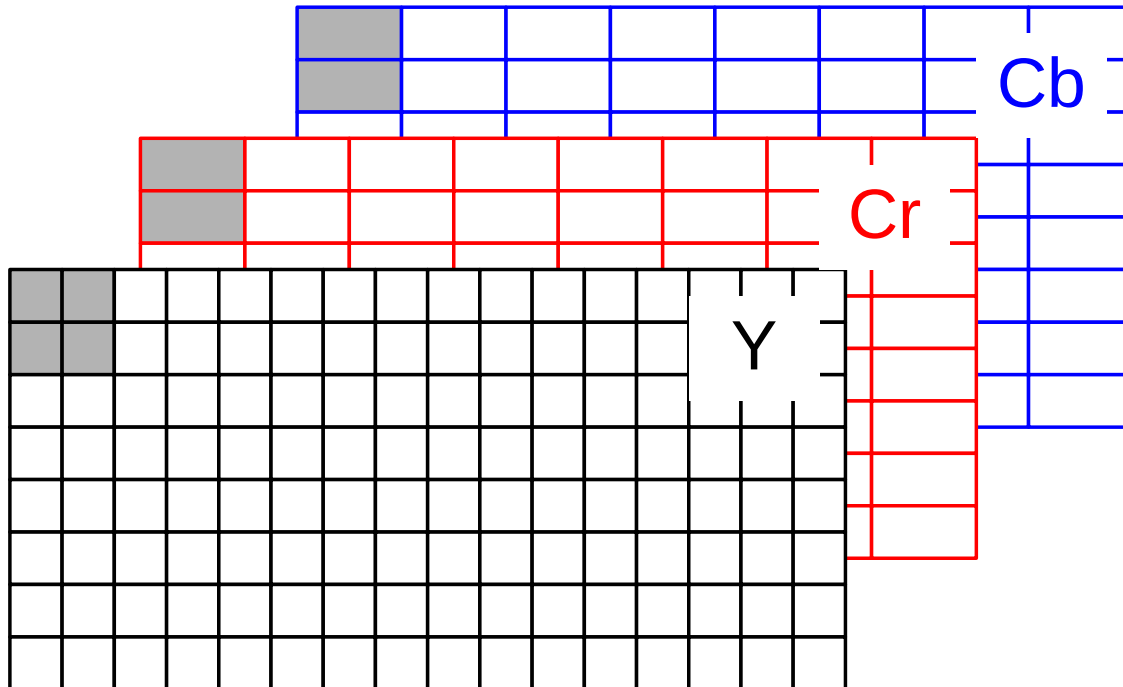


J:a:b notation

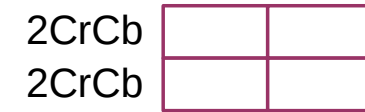
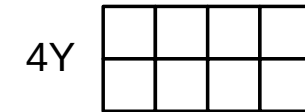


4:2:2 Video Sampling

4:2:2

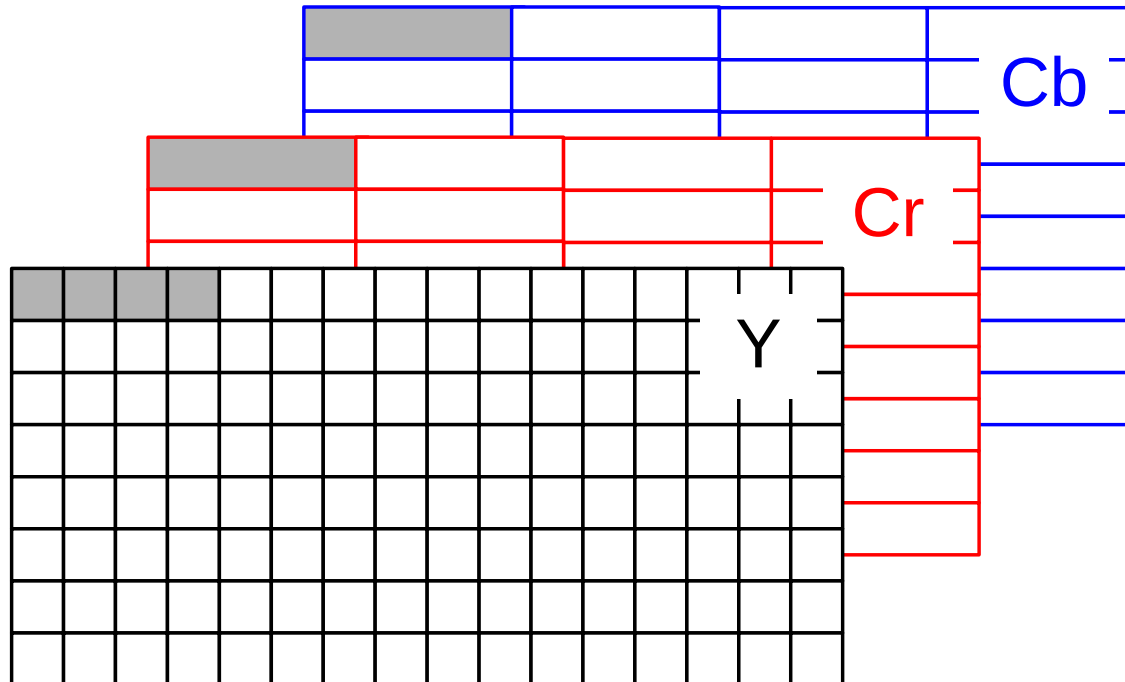


J:a:b notation

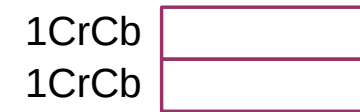
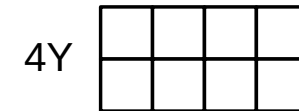


4:1:1 Video Sampling

4:1:1

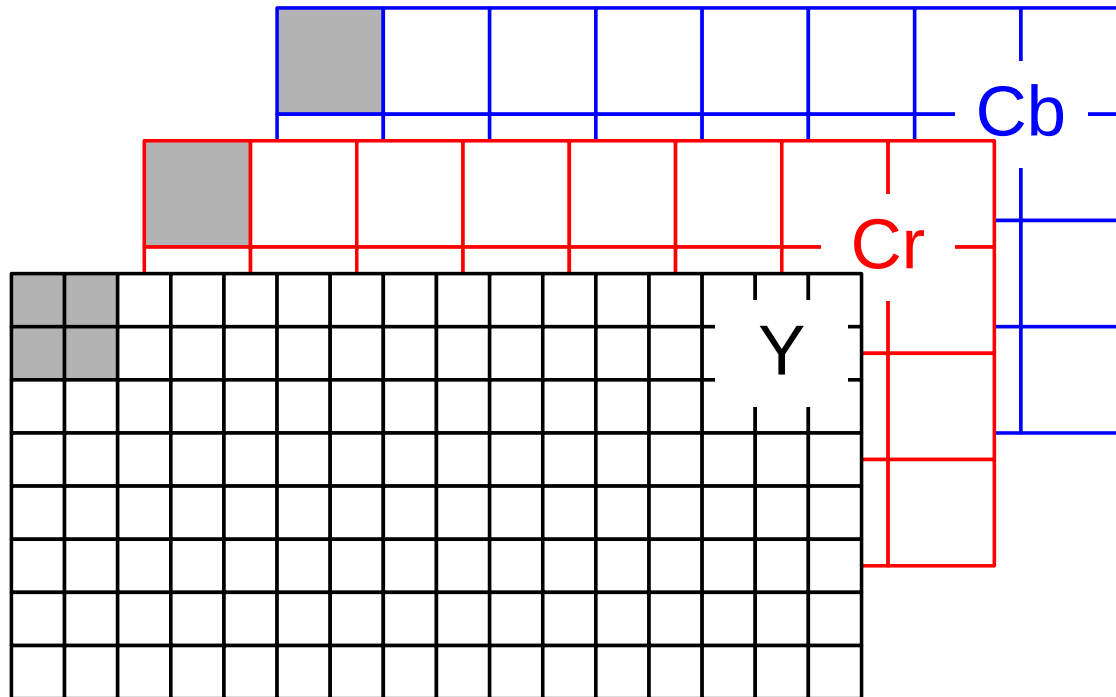


J:a:b notation

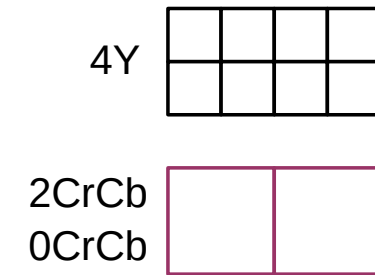


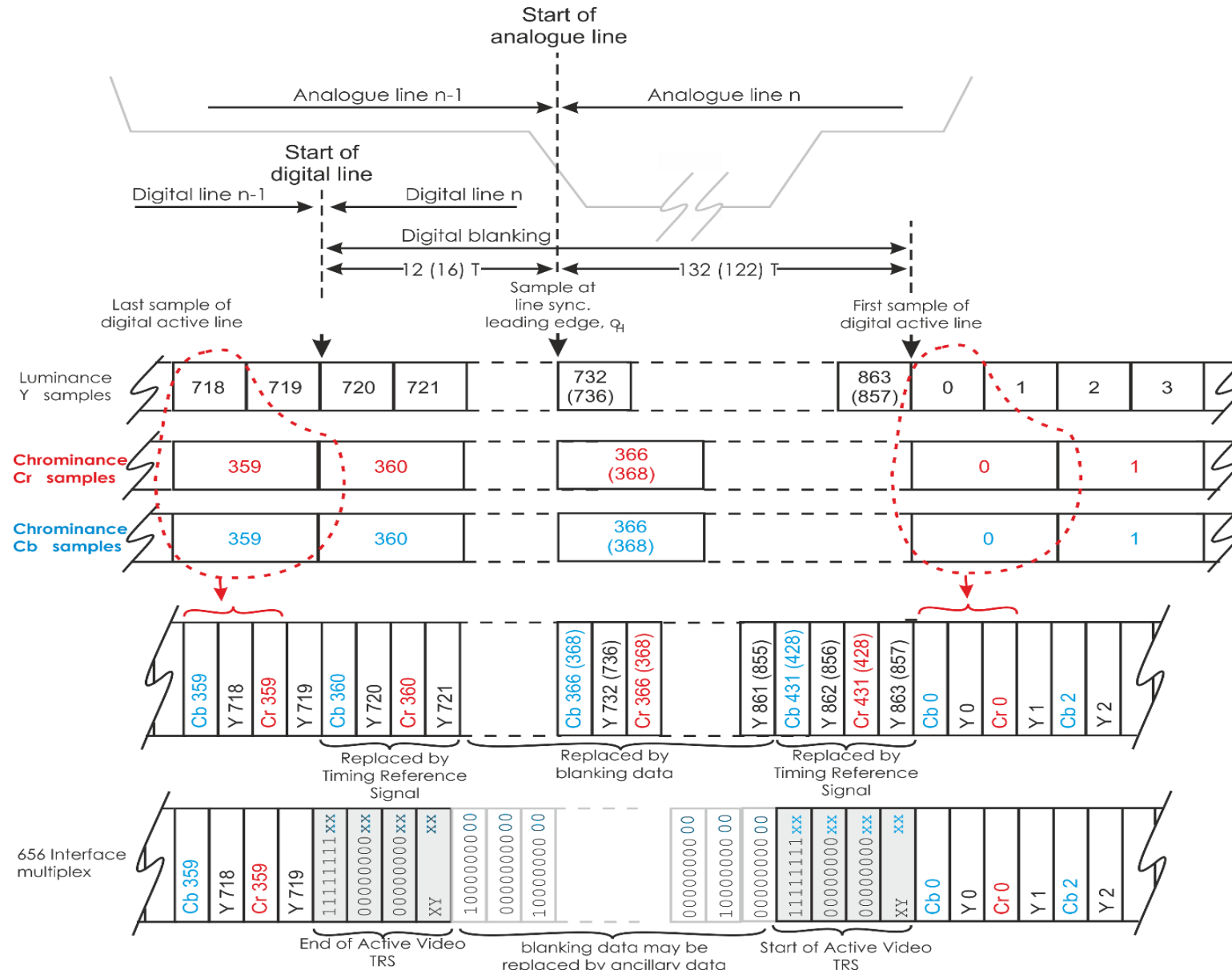
4:2:0 Sampling

4:2:0

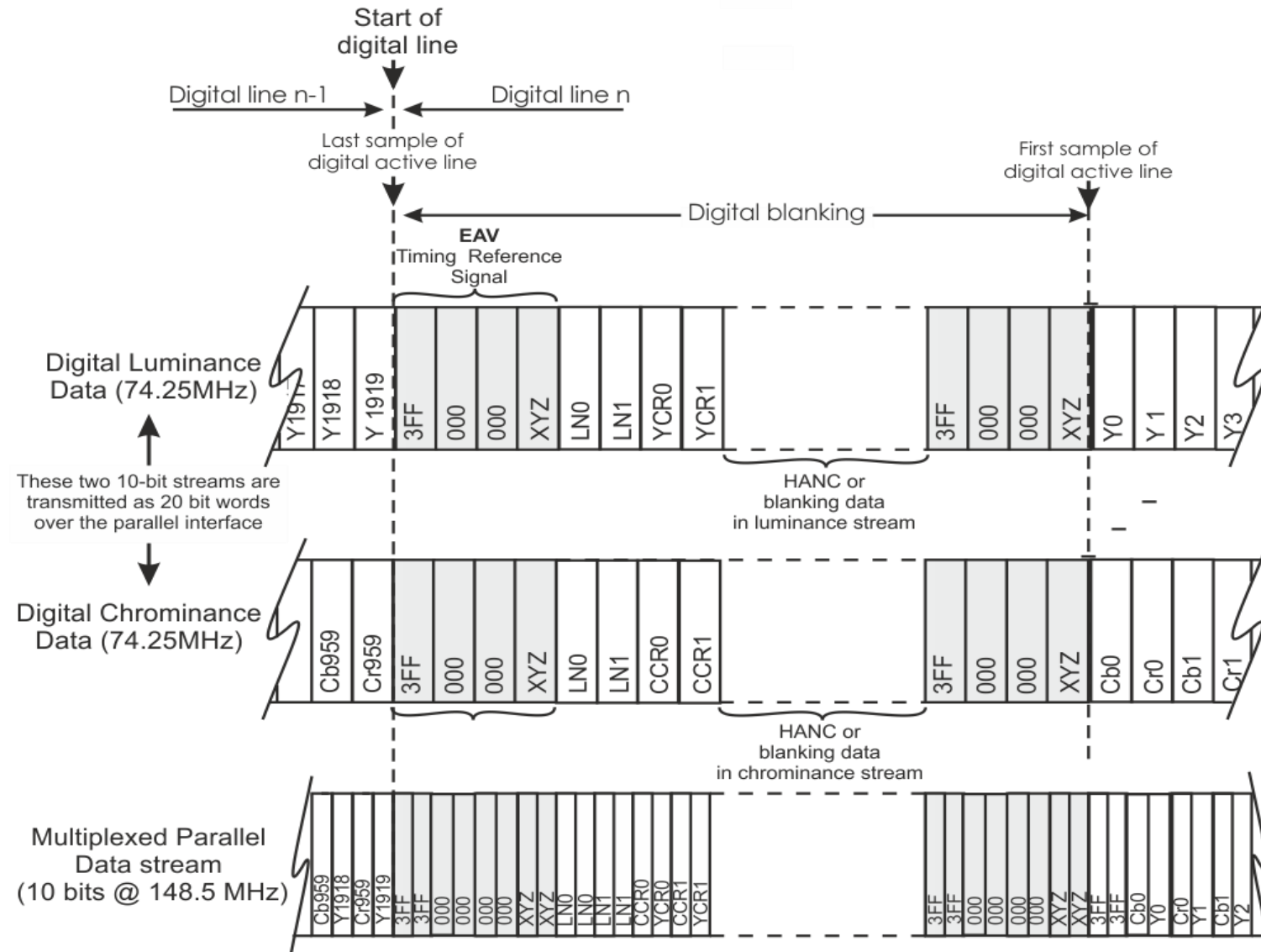


J:a:b notation

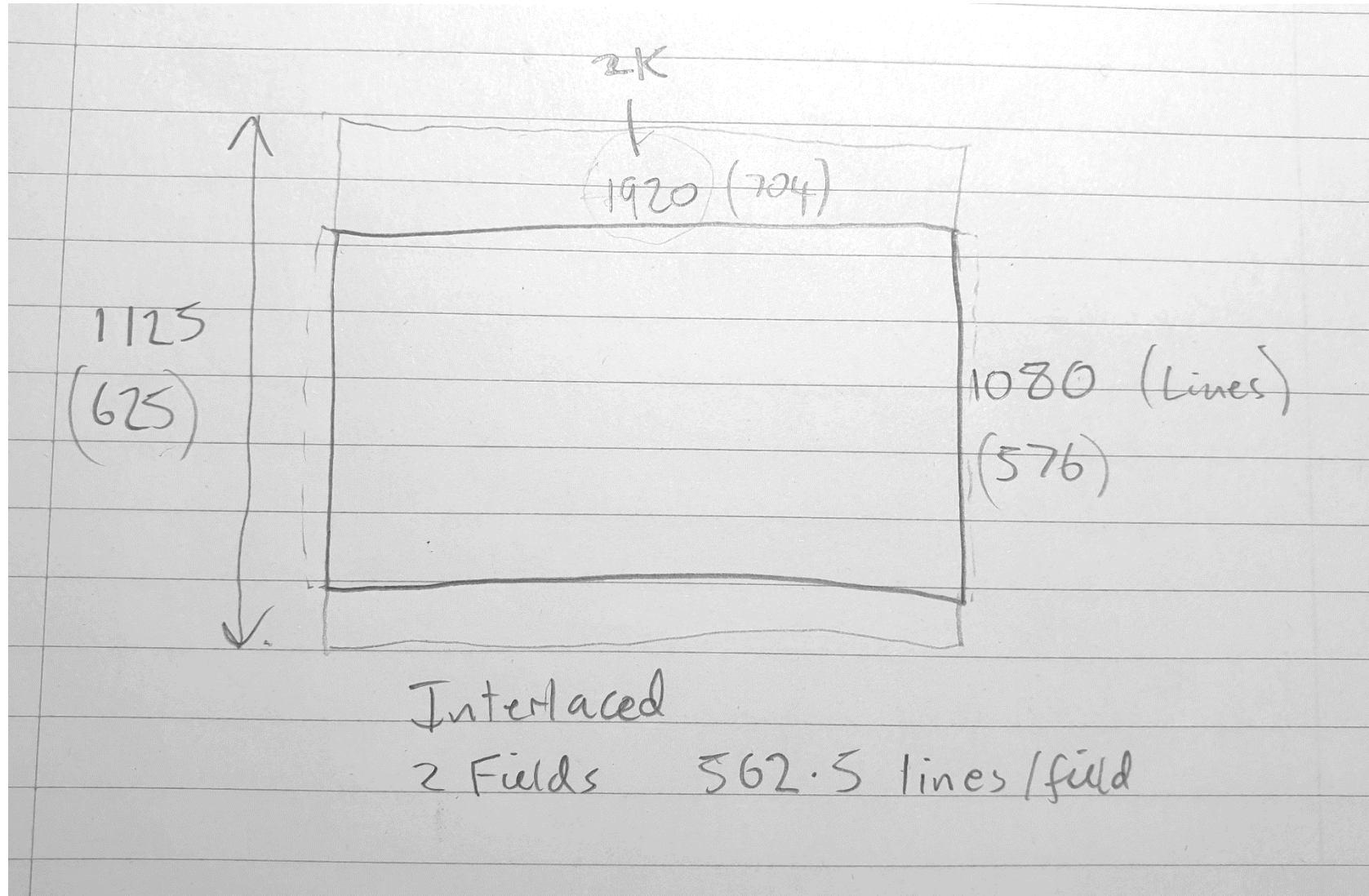


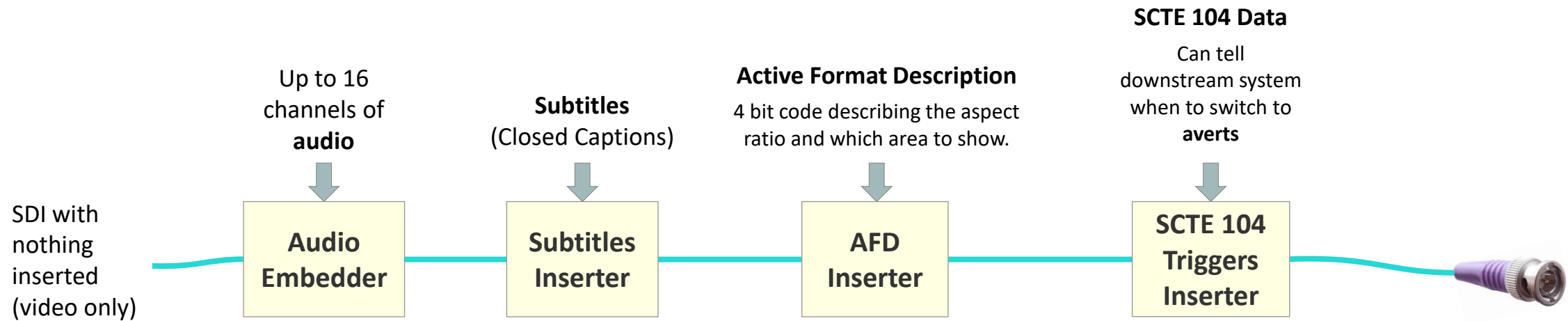


NOTE: Numbers in parenthesis () are for 525/60 systems
T = Luminance sample period.
The two least significant bits of the four word TRS are undefined (shown as xx above), though if generating the signal 11,00,00,00 (binary) are preferred.



NOTE: When the multiplexed parallel stream is serialised the bit rate clock is 1.485 Ghz
 LN0 and LN1 carry line number data
 YCR0 and YCR1 are luminance error check words
 CCR0 and CCR1 are chrominance check words





- **EDH** – Error Detection and Handling (for SD SDI only. CRC check is built into HD-SDI).
- **VPID** – Video Payload Identifier
 - Increasingly being used to describe the video format.
 - In early SDI it was possible to determine the video format by counting lines and samples
 - With the introduction of dual link interfaces, and segmented-frame standards, this is no longer possible;
 - SMPTE 352M
 - A list of the codes is also contained in the SMPTE Register:
 - <https://smpte-ra.org/video-payload-id-codes-serial-digital-interfaces>
- **Other SMPTE 291M Ancillary Data**
 - Timecode, closed captions, triggers (See SCTE 104), Lip sync data, control data, camera metadata etc...

- Ancillary data for SDI is defined in SMPTE ST 291M.
 - Each type has a 2 codes that identify it
 - DID Data Identifier
 - SDID Secondary Data Identifier
 - SMPTE maintains a register of the codes.
 - <https://smpte-ra.org/smpete-ancillary-data-smpte-st-291>

- Embedded Audio
 - 16 Channels for AES-3 data (including all extra bits – VUCP)
 - SMPTE ST 299-1

Standard	Name	Introduced	Common Bitrates	Example Video Formats
SMPTE 259M	SD-SDI	1989	270 Mbit/s	480/60i, 576/50i
SMPTE 292M	HD-SDI	1998	1.485 Gbit/s, and 1.485/1.001 Gbit/s	720/50p, 1080/25i 720/60p, 1080/30i
SMPTE 372M	Dual Link HD-SDI	2002	2.970 Gbit/s, and 2.970/1.001 Gbit/s	1080/50p, 1080/60p
SMPTE 424M	3G-SDI	2006	2.970 Gbit/s, and 2.970/1.001 Gbit/s	1080/50p, 1080/60p
SMPTE ST-2081	6G-SDI	2015	6 Gbit/s	2160/25p, 2160/30p
SMPTE ST-2082	12G-SDI	2015	12 Gbit/s	2160/50p, 2160/60p
SMPTE ST-2083	24G-SDI	tba	24 Gbit/s	2160/100p, 2160/120p 4320/50p, 4320/60p



- BNC (75Ω)

- Bayonet Neill–Concelman.
- Locking.
- Tool available for dense BNCs (e.g. on the rear of routers).

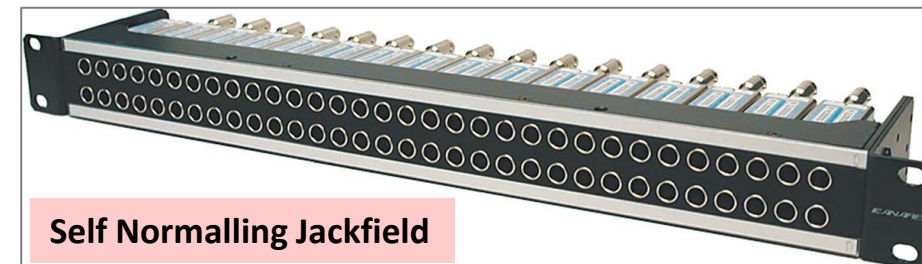
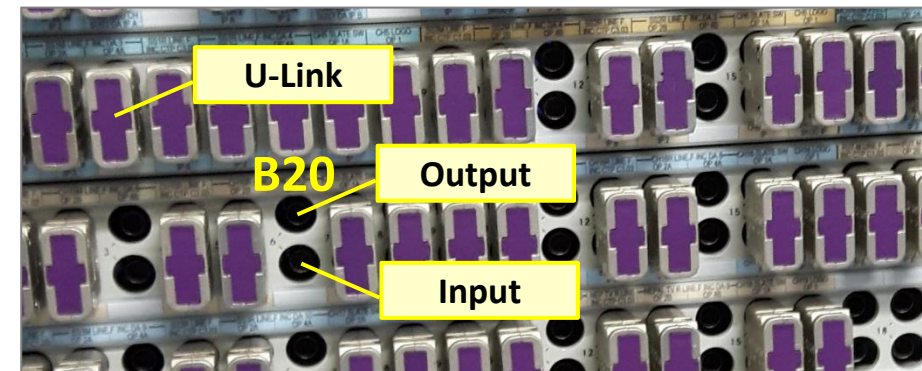
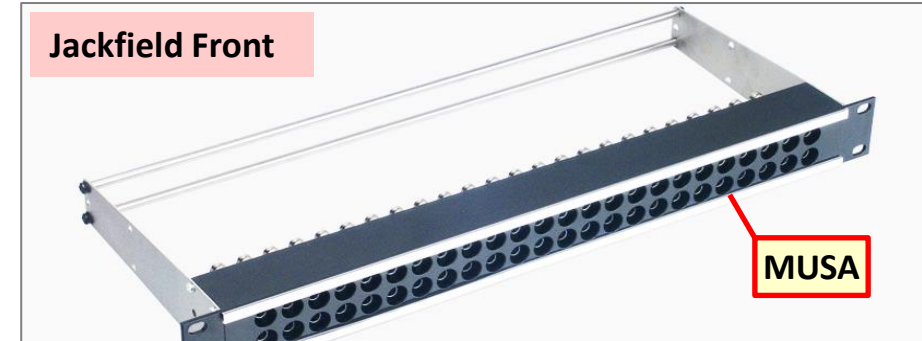
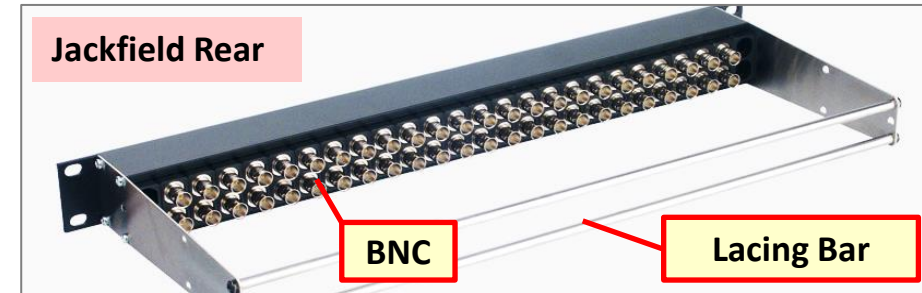
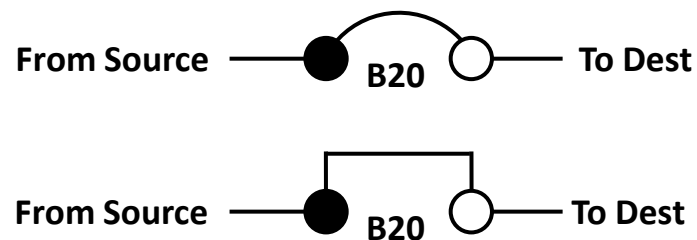


- MUSA

- Multiple Unit Steerable Array.
- Developed for RF aerial connections – Therefore originally 50Ω.
- Adopted for use in video jackfields (easy to slide in and out). Mismatch not critical for analogue video.
- Original MUSA not suitable for digital video – especially >3G SDI.
- New, reengineered versions are available which are 75Ω and therefore OK for SDI, up 3G (MUdigSA).

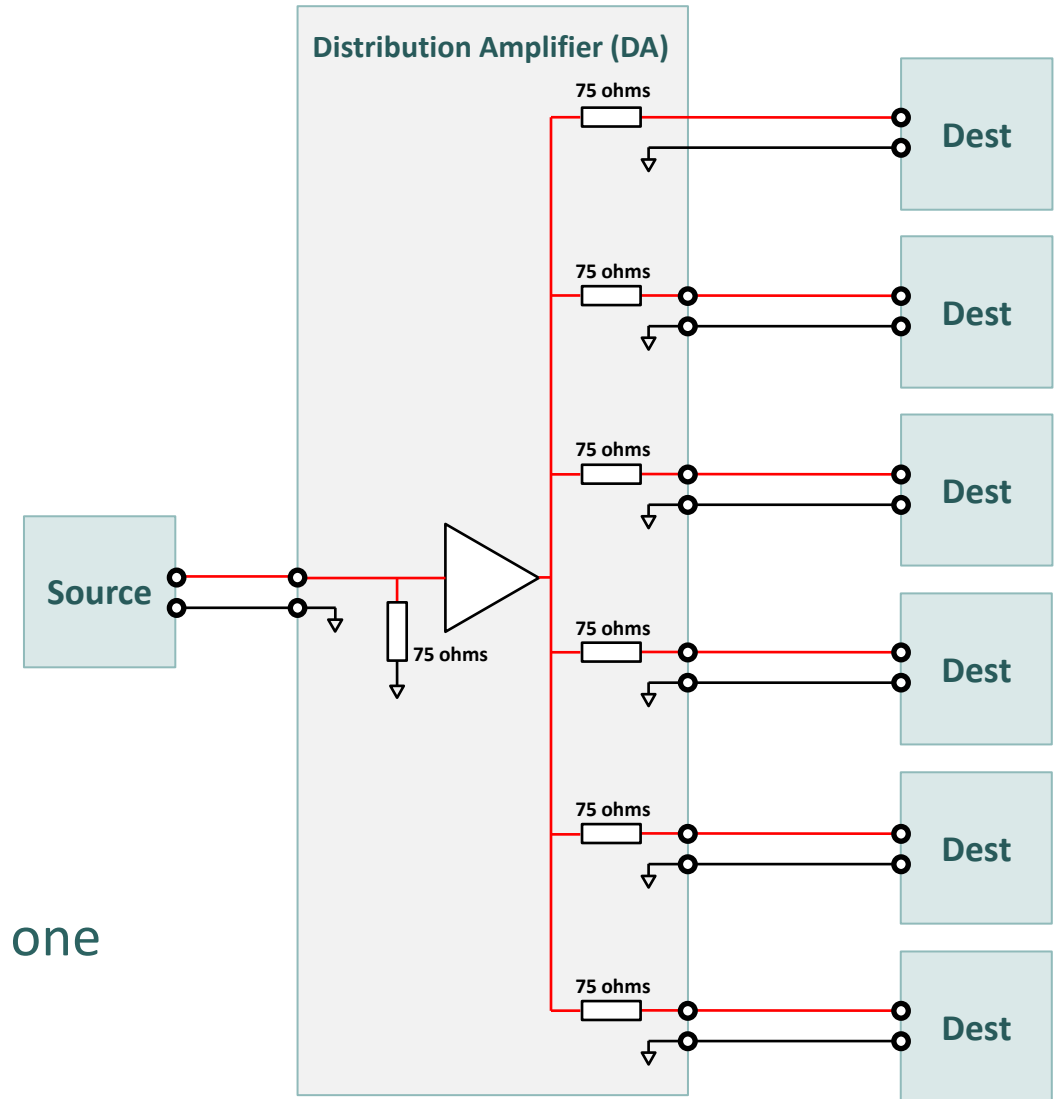


- Used to allow **breaking** of video signals
 - Occasional re-routing.
 - Accessing outputs for measuring / testing.
 - Injecting signals into inputs for testing.
- Jackfields are being **used less** as routers become cheaper
- **Normalled** connection
 - U-Link
 - Self Normalling (e.g. WECO)





- **SDI** and **ASI** use **coax cable** with **BNC connectors**.
- The cable must be terminated in 75 ohms.
- You **should not** split cables.
- If you need **multiple copies** of a signal and there is only one output available, you use a **Distribution Amplifier (DA)**.



DECS

DV CTP
108.A.

AXON DA
108.CH1.

Please ignore the strikethrough labels, they are not correct yet.

OP A1 / OP A2 – JF 108.G/I.1 – N/C / N/C (for monitoring only)

OP A3 > JF 108.E.1

OP B1 > N/C, OP B2 > N/C (temp cables to DCS21 114)

OP B3 > Evertz RTR IP 61 [078 FNSA]

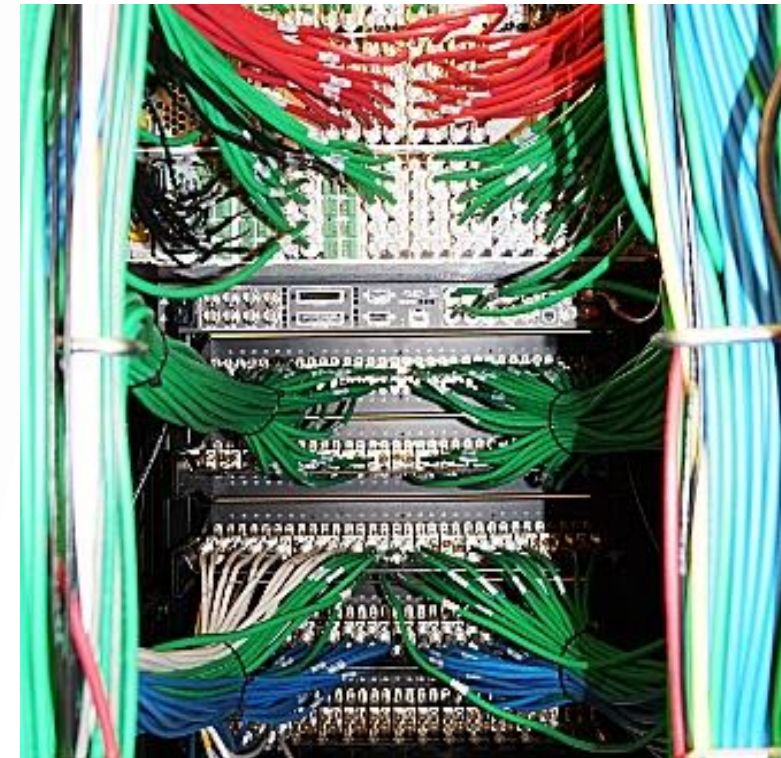
CTP = Central Termination Panel.

One of these:



1U 1x12 BNC

An DA made
by Axon



	Standard	259	292M	292M
	Name	SDI	HDSDI	3GSDI
	Bit rate Mbit/s	270	1485	2900
	Belden Cable			
Yellow	179DT	116m	34m	24m
	1855A	241m	79m	46m
Blue	1505F	261m	69m	
	1505A	338m	94m	67m
Green	1694A	436m	122m	82m
	7731A	622m	168m	110m

- **SMPTE 297-2006** defines an optical fibre system for transmitting **SDI** signals or **ASI**.
- Fibre connections are needed for HD-SDI connections of more than about 200m.
- This **not** IP / Ethernet. Pure SDI over fibre.
- When in optical form SDI signals can be split using a passive optical splitter instead of a distribution amplifier (DA).



The End